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Hermite Genocchi formula

If $\{x_0, \dots x_n\}$ are distinct nodes,
Then $f[x_0, x_1, \dots x_n] = \int\int_{\tau_n}$

$$\tau_n = \{(t_1, \dots t_n) \in \mathbb{R}^n / \sum_{i=1}^{n} t_i \le 1$$
$$t_i \ge 0\}$$

$$f^{(n)}(t_0 x_0 + t_1 x_1 + \dots t_n x_n)\, dt_1\, dt_2 \dots dt_n$$

$F: \mathbb{R}^{n+1} \to \mathbb{R}$ s.t.

$$F(x_0, x_1, \dots x_n) = \int_{\tau_n}\int f^{(n)}(t_0 x_0 + \dots + t_n x_n)\, dt_1 \dots dt_n$$

$\downarrow$

cont function

choose $x \in \mathbb{R}$ and $(x, \dots x) \in \mathbb{R}^{n+1}$

s.t. $\lim_{(x_0, x_1, \dots x_n) \to (x, x, \dots x)} F(x_0, x_1, \dots x_n) = F(x, x, \dots x)$

$$\| $$

$$f(x, x, \dots x)$$

$$f[x, \dots, x] = \frac{f^{(n)}(x)}{\lfloor n}$$

n^{th} components

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$$f[\underbrace{x}_{x_0}, \underbrace{x+h}_{x_1}, \underbrace{x+2h}_{x_2}, \dots \underbrace{x+nh}_{x_n}]$$

$$= \int_{T_n}\int f^{n}(t_0 x + t_1(x+h) + \cdots + t_n(x+nh))\, dt_1\, dt_2 \cdots dt_n$$

$$f(x, x+h, x+2h, \cdots x+nh) = \int_{T_n}\int f^{(n)}(t_0 + t_1 + \cdots + t_n)x + h(t_1 + 2t_2 + \cdots n t_n)\, dt_1\, dt_2 \cdots dt_n$$

$$f(x, x, \cdots x) = \lim_{h\to 0} f(x, x+h, \cdots, x+nh)$$

$$= \lim_{h\to 0} \int_{T_n}\int f^{n}(x + h(t_1 + 2t_2 + \cdots n t_n))\, dt_1\, dt_2 \cdots dt_n$$

$$= \int_{T_n}\int \lim_{h\to 0} f^{(n)}(x + h(___))\, dt_1 \cdots dt_n$$

$$= \int_{T_n}\int f^{(n)}(x)\, dt_1\, dt_2 \cdots dt_n$$

$$= f^{(n)}(x) \int_{T_n}\int dt_1 \cdots dt_n$$

$$= f^{n}(x)\, |T_n|$$

$$= \frac{f^{n}(x)}{\lfloor n} = f(x_0 \cdots x_0)$$

$f[x, x+h, x+2h, \ldots x+nh]$

$$p_n(x) = f(x_0) + \frac{f'(x_0)}{\lfloor 1}(x-x_0)$$

$$+ \frac{f''(x_0)}{\lfloor 2}(x-x_0)^2 + \cdots$$

$$\cdots + \frac{f^{(n)}(x_0)}{\lfloor n}(x-x_0)^n$$

let $x - x_0 = h$, $p_n(x)$

$= p_n(x_0 + x - x_0)$

$= p_n(x_0 + h)$

$$p_n(x_0+h) = f(x_0) + \frac{f'(x_0)h}{\lfloor 1} + \frac{f''(x_0)h^2}{\lfloor 2}$$

$$+ \cdots + \frac{f^{(n)}h^n}{\lfloor n}$$

$$|f(x_0+h) - p_n(x_0+h)|$$

$$= |R_n| \approx O(h^{n+1}) \longrightarrow 0$$

$|h| < 1$

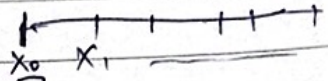
$$h_n = \frac{f^{(n+1)}(\xi_x)}{\lfloor n+1} h^{n+1}$$

Check

$\xi_x \in (x, x+h)$

$f[x, x+h, x+2h, \ldots x+nh]$

$= x \notin \{x_0, \ldots, x_n\}$



$f[x_0, x_1 \ldots x_n, x, x]$

$= \frac{d}{dx} f[x_0, x_1, \ldots x_n, x]$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$\frac{d}{dx} f[x_0, \ldots x_n, x]$

$$= \lim_{h \to 0} \frac{f[x_0, \ldots x_n, x+h] - f[x_0, \ldots, x_n, x]}{h}$$

$$f(x_0, x_1, x_2] = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$$

$$\lim_{h \to 0} \frac{f[x_0, \ldots, x_n, x+h] - f(x, x_0, x_1 \ldots x_n)}{x+h-x}$$

$$\lim_{h \to 0} \left[f[x, x_0, \ldots x_n, x+h] \right]$$

$$= f(x, x_0, \ldots x_n, x]$$

$$= f(x_0, x_1, \ldots x_n, x, x)$$